Interaction Strategies

Toward a Theory of Behavioral Organization

Foundational research paper

Abstract

Strategy is widely invoked to explain organized behavior, yet the term is used inconsistently across planning, reinforcement learning, decision theory, cognitive science, robotics, and human-computer interaction. This paper asks what an interaction strategy is, how it differs from action, plan, policy, heuristic, skill, and intention, and how it can be inferred without treating it as directly observed. The cross-disciplinary evidence rejects three reductions: a strategy is not a single action, not a mere observed sequence, and not necessarily an explicit plan. A stable synthesis emerges. An interaction strategy is an inferred, temporally extended organization of conditional action selection, information use, resource allocation, and contingency response that tends to preserve or improve one or more criteria across a class of interaction states. Strategies may be deliberative or learned, explicit or tacit, deterministic or stochastic, individual or distributed, and stable or adaptive. They are supported by traces but are not identical with traces. The paper proposes a typed strategy ontology, a policy representation separating goals, beliefs, action rules, information-seeking, commitments, and revision conditions, a lifecycle model, an inference protocol, and an evidence hierarchy. It concludes that existing theories are sufficient in their constituent mechanisms, but interaction research requires a shared strategy-analysis framework to connect behavioral traces, opportunity structures, transfer, and formal models.

1. Problem Statement

Interaction research often explains repeated or coordinated behavior by naming a strategy: users “search systematically,” “explore,” “satisfice,” “follow the interface,” “reuse a workflow,” or “recover by backtracking.” Such claims are stronger than descriptions of traces. A trace records what occurred under an observation model. A strategy claims that behavior was organized by a persistent or recurring structure that selected, sequenced, or revised actions across contingencies.

The central difficulty is evidential. The same trace may be generated by different strategies, and the same strategy may generate different traces when states, opportunities, costs, beliefs, or stochastic outcomes differ. A successful sequence does not prove advance planning. Repetition does not prove habit. A verbal explanation does not establish that the reported rule generated the observed behavior. Strategy must therefore be treated as a latent explanatory construct whose warrant depends on converging evidence and discriminating predictions.

The scientific task is not to introduce a new psychological substance. It is to identify the minimum structure required to distinguish strategy from neighboring concepts and to state how strategy claims can be represented, compared, tested, and falsified across human and artificial agents.

2. Review Method

The review is concept-centered rather than historical. Foundational sources were selected from philosophy of action and planning, human problem solving, reinforcement learning, decision theory, robotics, situated cognition, and human-computer interaction. Each tradition was analyzed for: (1) its unit of organization; (2) whether the organization is represented as a plan, policy, rule, procedure, intention, skill, or emergent pattern; (3) whether the construct is explicit or latent; (4) how it relates to goals and contingencies; (5) how it changes; and (6) what evidence identifies it.

The synthesis was accepted only where distinctions survived cross-domain counterexamples. Candidate definitions were rejected if they made every repeated behavior strategic, required conscious planning, collapsed strategy into policy, or could not distinguish a strategy from a post hoc description. The final framework inherits the prior research program’s separation between interaction episodes, behavioral traces, opportunity profiles, measurement claims, and transfer effects.

3. Existing Traditions

Planning theories treat plans as structures that organize future conduct. Bratman’s planning theory emphasizes that intentions and partial plans coordinate activity over time by constraining later deliberation and supporting stability, while allowing reconsideration. Cohen and Levesque formalize intention through commitment conditions, showing why a plan that is merely represented differs from one adopted for action.

Human problem-solving research represents organized behavior through problem spaces, operators, goals, subgoals, heuristics, and means-ends analysis. In this tradition, strategies are often higher-level regularities in search or problem decomposition rather than fixed action strings. Representation affects which strategies are available or efficient.

Reinforcement learning provides a precise policy concept: a mapping from states or observations to action distributions. Hierarchical reinforcement learning adds temporally extended options consisting of initiation conditions, internal policies, and termination conditions. These formalisms capture conditional action selection and temporal abstraction, but a complete interaction strategy may also include information-seeking, commitment, switching rules, resource constraints, and socially distributed coordination.

Decision theory explains action selection through preferences, beliefs, utilities, and constraints. Heuristics and bounded rationality explain organized departures from exhaustive optimization under limited time, information, and computation. A heuristic may be a component of a strategy, but strategies can coordinate multiple heuristics and include rules for choosing among them.

Situated-action research rejects the claim that plans fully determine conduct. Plans may orient, account for, or retrospectively render action intelligible, while actual conduct remains responsive to local contingencies. This critique does not eliminate strategy; it constrains strategy models to remain conditional, revisable, and situated.

Robotics separates high-level task planning from low-level control and increasingly integrates discrete plans with continuous motion, uncertainty, and risk. These systems demonstrate that strategic organization may span multiple representational and temporal levels.

HCI uses strategy to describe search, navigation, command use, error recovery, exploration, and expert procedure. GOMS and related cognitive models distinguish goals, operators, methods, and selection rules. Their strength is operational decomposition; their limitation is that they can overstate stability when strategies are inferred from sparse or highly constrained tasks.

4. Canonical Distinctions

Action is a bounded event or attributed control contribution. A strategy organizes the conditional selection and composition of actions across more than one possible state or contingency.

A plan is a prospective representation of intended organization. A strategy can be implemented without an explicit plan, and an explicit plan may never be adopted or executed.

A policy is a formal or functional mapping from state or observation to action distribution. A strategy may contain a policy, but often also includes state abstraction, information acquisition, commitment, stopping, switching, recovery, and revision rules.

A heuristic is a simplifying rule or procedure used under limited resources. It can be one component of a strategy.

A skill is a learned capacity for reliable performance. A strategy selects and coordinates skills; a skill can also embody strategic structure at a lower level.

A procedure specifies an ordered or conditional method. Strategies may invoke procedures but also decide when to depart from them.

An intention is a commitment-bearing practical attitude. Intention can stabilize a strategy, but strategies need not be conscious or intention-based.

A habit is a learned tendency triggered by context with reduced deliberative dependence. Habit may implement a strategy historically but can persist after the original criterion or context changes.

A tactic is a locally selected maneuver within a broader strategy. The distinction is relative to temporal and organizational scale.

A strategy is therefore not identified by explicitness, optimality, consciousness, success, or fixedness. Its defining role is organization across contingencies.

5. Canonical Definition

An interaction strategy is an inferred, temporally extended organization of conditional action selection, information use, resource allocation, and contingency response that tends to preserve or improve one or more criteria across a declared class of interaction states.

Each clause is necessary.

“Inferred” separates strategy from direct observation. “Temporally extended” excludes isolated actions. “Organization” requires relations among choices rather than mere recurrence. “Conditional” requires sensitivity to state, observation, belief, or contingency. “Information use” permits strategies whose primary function is search, diagnosis, or uncertainty reduction. “Resource allocation” includes time, effort, risk, attention, computation, and collaboration. “Contingency response” excludes rigid scripts that cannot handle variation unless rigidity itself is the declared strategy. “Criteria” avoids assuming a single explicit goal or utility. “Declared class of states” specifies the scope over which the strategy claim is intended to hold.

The definition is implementation-independent. It applies to humans, artificial agents, teams, and hybrid systems while allowing their internal mechanisms to differ.

6. Strategy Ontology

The minimum ontology contains the following entities and relations.

Agent or strategy-bearing system: the individual, artificial agent, team, or distributed arrangement to which the strategy is attributed.

Interaction domain: the class of situations and coupling structures within which the strategy is claimed to operate.

State and observation: the actual or estimated conditions relevant to selection. State need not be fully observable.

Opportunity profile: feasible, permitted, exposed, signaled, believed, and recoverable actions available under declared conditions.

Criterion set: goals, constraints, values, risks, deadlines, social commitments, or satisficing thresholds used to evaluate continuation or revision.

Action repertoire: primitive or temporally extended actions available to the strategy.

State abstraction: the distinctions the strategy treats as relevant. Two agents may face the same system state but organize behavior differently because they use different abstractions.

Selection rule: a deterministic, stochastic, heuristic, or deliberative relation connecting represented conditions to action.

Information policy: rules for observing, searching, querying, testing, or delegating before acting.

Commitment structure: conditions under which prior choices constrain later deliberation.

Contingency rule: responses to uncertainty, failure, interruption, or unanticipated state.

Switching and stopping rule: conditions for abandoning, changing, suspending, or completing the strategy.

Execution resources: time, attention, memory, computation, tools, collaborators, and authority.

Outcome and feedback: consequences that update beliefs, skills, habits, or future strategies.

Evidence base: records, traces, elicitation, interventions, and model comparisons supporting the attribution.

7. Strategy Taxonomy

Strategies should be typed along independent dimensions rather than placed in one exclusive hierarchy.

By control form: open-loop, closed-loop, receding-horizon, event-triggered, or mixed.

By representation: explicit symbolic plan, procedural rule, learned policy, heuristic bundle, reactive controller, or distributed coordination pattern.

By uncertainty handling: certainty-equivalent, information-seeking, risk-sensitive, robust, exploratory, or adaptive.

By temporal organization: immediate, episodic, hierarchical, persistent, or developmental.

By selection regime: deterministic, stochastic, satisficing, optimizing, lexicographic, threshold-based, or convention-based.

By revision: fixed, parameter-adaptive, structure-adaptive, opportunistic, or reflective.

By social organization: individual, delegated, negotiated, role-based, jointly committed, or emergent-distributed.

By learning relation: preconfigured, acquired, transferred, composed, imitated, or self-discovered.

By evidential status: hypothesized, weakly supported, discriminated from named rivals, causally supported, or formally verified for an artificial system.

8. Policy Representation

A reusable strategy representation is:

Sigma = <D, X, O, Z, C, A, pi, q, k, r, w, tau, rho, U>

D: declared interaction domain and scope.
X: relevant system and environment state space.
O: observation model.
Z: strategy’s state abstraction or belief representation.
C: criteria and constraints.
A: primitive and temporally extended action repertoire.
pi: conditional action-selection policy.
q: information-acquisition policy.
k: commitment and persistence rules.
r: recovery and contingency rules.
w: switching rules among substrategies.
tau: stopping and completion conditions.
rho: resource model.
U: uncertainty over model components and strategy attribution.

This representation does not assert that a human internally stores an explicit tuple. It is an analytical contract. Components may be implemented neurally, symbolically, socially, materially, or through artifacts.

A plan is a prospective representation generated or adopted under Sigma. A trace is a realization sampled from Sigma under particular states, observations, opportunities, disturbances, and outcomes. A policy is the pi component or, in broader formal usage, a subset of Sigma. A strategy instance is the activation of Sigma in a particular episode.

9. Strategy Grammar and Composition

A minimal grammar describes strategic organization without requiring one programming notation:

STRATEGY ::= ORIENT ; ORGANIZE ; ENACT ; MONITOR ; {ADAPT | RECOVER | SWITCH}* ; TERMINATE

ORIENT ::= establish criteria | classify situation | estimate state | identify constraints
ORGANIZE ::= select policy | construct plan | choose heuristic | allocate roles | reserve resources
ENACT ::= primitive action | option | procedure | delegated action | coordinated joint action
MONITOR ::= observe outcome | test assumption | track criterion | detect anomaly
ADAPT ::= update parameter | revise belief | refine plan | change abstraction
RECOVER ::= undo | compensate | retry | backtrack | seek assistance | replan

SWITCH ::= replace policy | invoke fallback | change goal priority | delegate
TERMINATE ::= criterion met | threshold reached | resource exhausted | infeasible | abandoned

Composition occurs through sequential composition, conditional branching, iteration, hierarchical invocation, parallel coordination, fallback, and arbitration. Composition is valid only when initiation conditions, resource assumptions, authority, and termination semantics are compatible.

Strategy equivalence is typed. Two strategies may be outcome-equivalent but differ in risk, information use, opportunity dependence, trace distribution, recoverability, or transfer. No universal strategy-equivalence relation is justified.

10. Strategy Lifecycle

The lifecycle is recurrent rather than strictly linear.

1. Formation: a strategy is generated, selected, learned, inherited, or socially assigned.
2. Adoption: the strategy becomes behaviorally controlling or commitment-bearing.
3. Initialization: state, criteria, opportunity, and resources are assessed.
4. Enactment: actions and information-gathering operations are selected.
5. Monitoring: outcomes, deviations, and uncertainty are evaluated.
6. Adaptation: parameters, beliefs, plans, or action rules change while strategy identity is retained.
7. Switching: a materially different organizing structure replaces or suspends the current one.
8. Recovery: the agent repairs consequences or returns to a viable state.
9. Termination: completion, abandonment, interruption, or supersession occurs.
10. Consolidation: experience updates skill, habit, memory, artifacts, or reusable strategy components.
11. Transfer: components are activated or transformed in a new context.

Identity across adaptation requires declared invariants. A strategy may remain the same if its criterion structure, state abstraction, or high-level policy remains stable while parameters change. If the organizing rule or criteria change beyond the declared tolerance, the model should record a strategy switch rather than adaptation.

11. Inferring Strategies

Strategy inference is an abductive and model-comparison problem. It should not begin with labels assigned to visually similar traces.

The inference protocol is:

1. Declare the episode boundaries, observation model, and behavioral trace.
2. Specify the strategy-bearing system and level of attribution.
3. Enumerate feasible rival strategies before inspecting outcomes where possible.
4. For each candidate, state predicted action distributions, information requests, latency patterns, error responses, switching behavior, and outcome distributions across diagnostic contingencies.
5. Collect repeated or strategically informative episodes rather than relying on one successful sequence.
6. Use interventions that separate rivals: change information availability, action cost, risk, feedback, state structure, or opportunity.
7. Incorporate elicited plans and verbal reports as fallible evidence, not privileged access.
8. Fit candidate models with uncertainty and penalize unnecessary complexity.
9. Evaluate out-of-sample prediction, calibration, and counterfactual discrimination.
10. Report underdetermination where multiple candidates remain observationally equivalent.

Strong strategy evidence comes from cross-context stability plus diagnostic adaptation. A candidate should predict not merely common actions but different responses under contingencies where rival strategies diverge.

12. Strategy Evidence Hierarchy

Level 0 - Raw record: timestamps, sensor values, commands, utterances, or system events.

Level 1 - Verified behavioral event: a coded event supported by an explicit detection rule.

Level 2 - Episode trace: ordered or partially ordered attributed events under declared boundaries.

Level 3 - Descriptive regularity: recurrence, sequence motif, transition probability, or path distribution.

Level 4 - Candidate strategic organization: a model connecting state, observation, criteria, and conditional selection.

Level 5 - Rival discrimination: evidence that the candidate explains diagnostic contingencies better than specified alternatives.

Level 6 - Transfer and adaptation evidence: the candidate predicts persistence, modification, or reuse across contexts.

Level 7 - Causal support: intervention on a proposed strategy component changes behavior as predicted.

Level 8 - Mechanistic or implementation support: independent evidence identifies the process or architecture implementing the strategy.

Level 9 - Formal verification: for declared artificial systems, the implemented policy and switching structure are proven to satisfy specified properties.

Claims must not skip levels. A repeated sequence supports a regularity, not by itself a strategy. A verbal description supports reported planning, not necessarily the generative organization of action.

13. Comparative Cases

Novice and expert. A novice may use serial visible search and local trial-and-error; an expert may use recognition, hidden commands, hierarchical chunks, and anticipatory recovery. Equal success does not imply equal strategy. Evidence must distinguish knowledge, motor fluency, opportunity belief, and state abstraction.

Deterministic and adaptive policies. A deterministic scripted agent can instantiate a strategy when the script includes conditional organization across states. An adaptive policy may preserve strategy identity while updating parameters. Stochasticity does not negate strategy; randomness may implement exploration or robustness.

Human and artificial agents. Artificial policies may be inspectable at implementation level, while human strategies usually remain inferred. Formal equivalence between a human model and an artificial policy does not imply psychological identity.

Individual and collaborative strategies. Team strategy may reside in role allocation, communication protocol, shared artifacts, and mutual commitments rather than in any one participant. A distributed strategy can persist despite membership changes if coordination invariants remain stable.

Situated improvisation. Locally improvised action can still be strategic when it preserves criteria through conditional organization. Suchman’s critique rules out treating the plan as exhaustive cause, not the possibility of strategic structure.

Interface migration. A user may preserve a high-level strategy while replacing low-level actions, or preserve a learned sequence while losing its effectiveness. Transfer should therefore be assessed component by component.

14. Strategy, Transfer, and Learning

Transfer is not proof of strategy identity. A source strategy may transfer intact, only a heuristic may transfer, or a procedure may be reused under an incompatible state abstraction. Positive performance transfer can occur through increased familiarity without preservation of the original strategy.

Strategy transfer requires a correspondence between source and target components: criteria, state abstractions, opportunity structures, action repertoires, and contingency rules. The strongest evidence shows that a source-derived organizing rule predicts target behavior under diagnostic conditions better than alternatives.

Strategy evolution can occur through parameter tuning, policy refinement, composition, decomposition, abstraction change, habit formation, externalization into tools, social redistribution, or replacement. Researchers should distinguish within-strategy adaptation from strategy switching using predeclared identity criteria.

15. Falsification and Rival Accounts

The framework should be narrowed or rejected under five conditions.

Policy sufficiency: if a conventional policy mapping plus state representation and reward fully captures every relevant strategic distinction, the broader strategy contract is unnecessary. The present analysis finds that information policies, commitments, switching, recovery, resources, and distributed organization often require explicit treatment, but these could be incorporated into an augmented policy formalism.

Trace sufficiency: if strategy labels add no predictive or explanatory gain beyond trace features and context, strategy should remain a descriptive category rather than a latent construct.

Plan identity: if explicit plans consistently and exhaustively determine behavior across contingencies, situated revision components are unnecessary. Existing evidence strongly contradicts this universal claim.

Inference unreliability: if independent analyses cannot recover comparable strategy attributions or if rival models remain observationally indistinguishable, claims must be restricted to descriptive regularities.

No transfer or intervention gain: if inferred strategies do not predict adaptation, transfer, or responses to diagnostic interventions better than simpler models, strategy attribution lacks scientific value.

16. Threats to Validity

Strategy is a scale-relative concept. What appears strategic at one level may be a primitive action at another. Every analysis must declare temporal scale and unit of attribution.

Human strategy inference is vulnerable to post hoc rationalization, observer bias, and flexible labeling. Artificial-agent strategy descriptions are vulnerable to anthropomorphism in the opposite direction: an optimized policy need not contain human-like goals, intentions, or reasoning.

Outcome-based selection creates circularity when a “successful strategy” is defined by success and then used to explain it. Criteria and candidate structures should be specified independently of observed outcomes.

Sparse observations may make different strategies observationally equivalent. Missing information about beliefs, opportunity, and external artifacts can misattribute distributed organization to the individual.

Cultural, institutional, and collaborative strategies may not decompose cleanly into individual policies. The framework permits distributed attribution but does not solve every boundary problem.

17. Implications for Interaction Research

The strategy framework connects the prior theory stack without collapsing its layers. Interaction defines reciprocal or recurrent coupling. Behavior provides attributed trajectories. Measurement specifies how records support claims. Similarity supplies typed comparison. Transfer identifies source-dependent effects across contexts. Strategy explains conditional organization across actions and contingencies.

For empirical research, strategy should be modeled only after opportunity and observation conditions are represented. An action absent from the believed opportunity set cannot be treated as deliberately rejected. A failed recovery sequence may reflect missing permission or inaccessible signaling rather than a poor strategy.

For formal modeling, the strategy contract identifies requirements that a candidate formalism must support: partial observability, temporal abstraction, information action, resource limits, switching, recovery, composition, distributed control, and uncertainty over attribution.

For design and evaluation, the framework discourages normative shortcuts. A strategy that is slower may be more robust, auditable, transferable, or accessible. Comparison must therefore remain criterion-relative and typed.

18. Conclusion

Existing theories are sufficient in their constituent concepts, but no single inherited term cleanly coordinates strategy across planning, reinforcement learning, human problem solving, situated action, robotics, and interaction research.

The warranted synthesis is not a new psychological primitive. It is a strategy-analysis contract that treats strategy as an inferred, temporally extended organization of conditional action selection, information use, resource allocation, and contingency response. Strategies are supported by traces but are not traces; may include plans but are not identical with plans; may be represented as policies but often require information, commitment, switching, recovery, and resource structure; and may transfer only partially across contexts.

Scientific strategy claims require rival models, diagnostic contingencies, explicit uncertainty, and evidence beyond successful action sequences. This framework supplies the bridge from behavior and transfer to the next phase of the research program: determining which formal models can represent interaction episodes, opportunity profiles, traces, strategies, composition, and uncertainty without losing the distinctions already established.

19. Reusable Artifacts

A. Strategy Ontology

Element	Definition
Strategy-bearing system	Individual, artificial agent, team, or distributed arrangement to which organization is attributed.
Domain	Declared class of interaction situations.
State/observation	Actual and accessible conditions relevant to selection.
Opportunity profile	Typed action availability and recoverability.
Criteria	Goals, constraints, values, risks, commitments, or thresholds.
State abstraction	Distinctions treated as decision-relevant.
Action repertoire	Primitive and temporally extended actions.
Selection policy	Conditional relation from represented conditions to action.
Information policy	Rules for observing, searching, testing, or querying.
Commitment	Conditions that stabilize prior choices.
Contingency/recovery	Rules for failure, uncertainty, and repair.
Switch/stop rules	Conditions for replacement, suspension, completion, or abandonment.
Resources	Time, attention, memory, computation, tools, authority, and collaborators.
Evidence	Records and tests supporting attribution.

B. Strategy Taxonomy

Dimension	Types
Control	Open-loop; closed-loop; receding-horizon; event-triggered; mixed
Representation	Plan; procedure; policy; heuristic bundle; reactive controller; distributed pattern

Uncertainty	Certainty-equivalent; information-seeking; risk-sensitive; robust; exploratory; adaptive
Selection	Deterministic; stochastic; satisficing; optimizing; lexicographic; threshold; convention
Revision	Fixed; parameter-adaptive; structure-adaptive; opportunistic; reflective
Social	Individual; delegated; negotiated; role-based; jointly committed; distributed
Learning	Preconfigured; acquired; transferred; composed; imitated; self-discovered
Evidence	Hypothesized; supported; rival-discriminated; causally supported; verified

C. Strategy Contract

- Declared domain and attribution level
- Episode boundaries and timescale
- Criteria and constraints
- State and observation model
- State abstraction or belief model
- Opportunity assumptions
- Action and skill repertoire
- Conditional selection policy
- Information-acquisition policy
- Commitment and persistence rules
- Contingency and recovery rules
- Switching and stopping conditions
- Resource and authority assumptions
- Expected outcomes and risks
- Identity invariants under adaptation
- Evidence and rival strategies
- Uncertainty and validity domain

D. Strategy Inference Protocol

1. Declare episodes, traces, and observation model.

2. Specify strategy bearer and temporal scale.

3. Enumerate rival strategies.

4. Derive diagnostic predictions for each candidate.

5. Collect repeated and diagnostic episodes.

6. Intervene on information, cost, risk, feedback, or opportunity.

7. Treat reports and plans as fallible evidence.

8. Fit models with uncertainty and complexity control.

9. Evaluate held-out and counterfactual discrimination.

10. Report underdetermination and prohibited interpretations.

E. Strategy Evidence Hierarchy

Level	Evidence status
0	Raw record

1	Verified behavioral event
2	Episode trace
3	Descriptive regularity
4	Candidate strategic organization
5	Rival discrimination
6	Transfer/adaptation support
7	Causal intervention support
8	Mechanistic implementation support
9	Formal verification

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